www.tntextbooks.in

Solution

(i) Torque about point D is zero. (as F passes through D).

Torque about point C is zero. (as F passes through C).

Torque about point O is $(\overrightarrow{OC})\times\vec{F}$ and direction is along GH.

Torque about point B is $(\overrightarrow{BD})\times\vec{F}$ and direction is along GH

(The $\perp$ of $\overrightarrow{BD}$ with respect to $\vec{F}$ is $\overrightarrow{OC}$).

(ii) Torque about axis CD is zero (as F is parallel to CD).

Torque about axis OC is zero (as F intersects OC).

Torque about axis AB is zero (as F is parallel to AB).

Torque about axis GH is $(\overrightarrow{OC})\times\vec{F}$ and direction is along GH.

The torque of a force about an axis is independent of the choice of the origin as long as it is chosen on that axis itself. This can be shown as below.

Let O be the origin on the axis AB, which is the rotational axis of a rigid body. F is the force acting at the point P. Now, choose another point O' anywhere on the axis as shown in Figure 5.10.

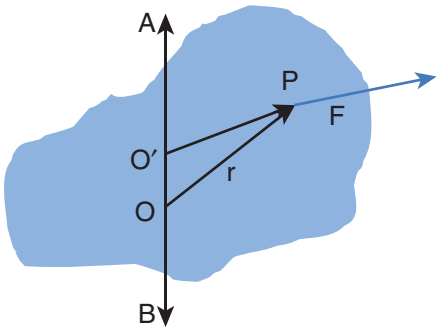


Figure 5.10. Torque about an axis is independent of origin



Identify the direction of torque in country press shown in picture (in Tamil, 'Marasekku' மரச்செக்கு)